Using Temporal Fusion Transformers for Market Regime Detection

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1 Abstract

This paper aims to explore the use of a time series-based transformer model known as the Temporal Fusion Transformer (TFT) model for detecting market regimes. It utilizes different data types to attempt to enhance the predictive power of the TFT and compares to a Long-Short Term Memory model (LSTM). Current popular models of market regime detection include Hidden Markov Models (HMMs) and Markov-Switching Autoregressive Models (MSARs), among other unsupervised machine learning techniques such as K-means clustering. Enhancing the predictive power of deep neural networks (DNNs) such as the TFT and tuning it for classification of regimes can improve strategy development and allow for dynamic adjustment in strategy deployment. This study ultimately concluded that at the set parameters and data set given, the Temporal Fusion Transformer was not able to outperform the predictability of the LSTM, however, it did show potential for detecting shifts in data structure.

2 Introduction & Literature Review

Financial markets are largely dynamic systems that can often be categorized into certain regimes characterized by sustained transformations. Regime changes in financial markets refer to drastic changes in overall market behavior causing changes in market conditions such as volatility and asset prices. Such large shifts can be attributed to few but largely influential factors including changes in regulation and policy or other major external events such as influential bankruptcies (Ang & Timmermann, 2011; Chen & Tsang, 2020). As such, these periods of structural shifts result in changes in statistical properties of markets. Periods of high volatility often correspond with decreases in economic performance – bear markets – while periods of low volatility often indicate increases in economic performance – bull markets (Ang & Timmermann, 2011). Thus, regime detection is crucial for understanding the dynamics of financial markets and can lead to better risk management strategies, capital allocation, and strategy development.

Regimes detection methods vary largely in terms of their classification. Probabilistic methods including HMMs and MS-ARs were adapted for the purpose of uncovering hidden states within the time series. The MS-AR model extends on the work of classic the Markov chain and assumes next state only depends on the current state without considering previous states. Through Hamilton's model, recessionary periods were able to be clearly identified using the National Bureau of Economic Research (NBER) as a benchmark (1989). MS-AR is ideal for capturing asymmetries because of its ability to dynamically switch parameters – mean, variance, etc. – of its calculation with each new state (1989).

Hidden Markov Models are another type of regime detection method which utilizes unsupervised machine learning to identify hidden states within time series. Basic HMMs use time series data. Parameters of HMMs include a transition probability matrix, an emission probability matrix, and an initial state distribution. The transition probability matrix is calculated to give the likelihood of switching from one hidden state to another. The emission probability distribution tells that probability of a certain output occurring given a hidden state. Lastly, the initial state distribution is the initial probability of the Markov chain. It then utilizes an Expectation-Maximization Algorithms, such as Baum-Welch, to learn and optimize the parameters. Initial estimates are computed for probabilities and iteratively improved. Iterations occurs until optimal transition probabilities and emission probabilities are found.

Another regime detection method is known as k-means clustering. This technique provides and unsupervised machine learning approach to classifying patterns within input data to a set of input clusters. The clusters are classified based on derived statistical characteristics such as their mean and variance. The optimal number of clusters can often be determined by Knee location algorithms that mathematically solve for points of inflection within a plot that by testing for different numbers of clusters. When used for regime detection, input volatility data clusters may often be representative of market regimes such as bull, bear, or stationary periods (McGreevy, 2023).

Traditional neural networks such as Recurrent Neural Networks (RNNs) and derived from them Long Short-Term Memory (LSTMs) used for forecasting do not perform nearly as well as TFTs for forecasting (Hartanto & Gunawan, 2024; Frank, 2023). This is because of their weaknesses in utilizing long-range dependencies given their structure. The Transformer architecture solves this problem by introduction encoder, decoder, and gated attention mechanisms which allow networks to remember important information. Furthermore, the development of Transformers for time series forecasting has shown promising results, especially for time series forecasting. In one study, conducted by Mozaffari & Zhang, they conduct a study in which they forecast closing prices using an LSTM, Prophet Model and Transformer model. The Transformer provided the best predictive accuracy for forecasting future closing prices of equites incident American Airlines Group and Atlantic American-Life Insurance with its metrics consistently scoring lower on the Test and Validation sets (2024). Furthermore, there have been models derived from the Transformer architecture of such as Informer, provide improved performance of the base transformer by reducing time complexity and allowing the model to highlight important sequence components (Zhou et al., 2020). These improvements can provide a baseline for potentially detecting regimes by allowing for detections of structural changes in complex time series data.

The Temporal Fusion Transformer (TFT) is a model inspired by the by the transformer architecture and is specifically utilized for multi-horizon forecasting. What makes this model special is its ability to provided multiple predictions into the future. Components of this model include Gated Attention Mechanisms, Variable Selection Networks, Static Covariate Encoders, Temporal Processing, and Prediction Intervals (Lim et al., 2019). It can utilize these variables to output interpretable multi-horizon forecasts. This research aims to develop and train a TFT and compare its performance of regime detection against traditional methods such as HMMs and MS-ARs.

3 Methodology

This study integrated high frequency data and low frequency data to detect and forecast regimes. It integrates dataset utilized by Frank (2023) to forecast realized volatility.

Data Sources

Daily price data and volume data for the SPY Exchange Traded Fund (ETF) were retrieved from the yfinance API (not affiliated with Yahoo! Finance). Macroeconomic data was retrieved from the Federal Reserve Economic Data (FRED) API. Lastly data for economic uncertainty was retrieved in csv format via the Economic Policy Uncertainty website. All data can be summarized here:

Name	Source	Frequency
S&P 500 Index	*yfinance API*	*daily*
SPY ETF	*yfinance API*	*daily*
CBOE VIX	*FRED API*	*daily*
Moody's Seasoned AAA Corporate Bond Yield	*FRED API*	*daily*
ICE BofA US High Yield Index Option-Adjusted Spread	*FRED API*	*daily*
Federal Funds Effective Rate	*FRED API*	*monthly*
Market Yield on U.S. Treasury Securities at 30-Year Constant Maturity	*FRED API*	*monthly*
Sector Employment(various)	*FRED API*	*monthly*
Uncertainty Index	*Economic Uncertainty Policy Website (downloaded as csv)*	*monthly*

Table 1. Data names, sources, and frequency

Data Transformation

To ensure all-time series data was synced, they were all aligned to daily frequency based on the largest frequency data set. In this case it was CBOE VIX. Macroeconomic indicators tend to have lower frequency data (quarterly, monthly), therefore, in an effort harmonize the data, they were extended and forward filled to match with VIX data frame. Furthermore, a new column title '{macroeconomic indicator} _update_flag' was introduced to indicate updates in values for each extended data frame. For high frequency indicators with minimal missing values, missing data points were interpolated. NaN values were dropped if they were at the beginning of the dataset. After all transformations, the data set included daily data from 2000-01-03 to 2021-12-31. The training data split occurred from 2000-01-03 to 2018-12-31. The validation data split ranged from 2019-01-01 to 2021-12-31. The final dataset had 5,739 rows of data. This date range was used to ensure coverage of clear market regime shifts including 2008 Market Crash and COVID.

Feature Engineering

Deriving features from given datasets is important as it can give the model new context for better predictions. Feature engineering for this data set can be summarized as follows:

Feature Name	Source
SPY Logarithmic Returns	SPY Close
Realized Volatility (RV)	SPY Logarithmic Returns
Realized Volatility 1-day lag	SPY Realized Volatility
Realized Volatility 2-day lag	SPY Realized Volatility
Realized Volatility 3-day lag	SPY Realized Volatility
Squared Log Return	SPY Logarithmic Returns

Table 2. Feature engineering with source and new feature.

Regime Labeling Strategy

Traditional models such as HMMs or MS-ARs usually compare two types of regimes: those with high volatility and those with low volatility. For the purposes of this model, an unsupervised K-means clustering technique was used to classify regimes based on realized volatility. To identify the optimal number of clusters, the Elbow Technique Algorithm was deployed. This method allowed us to obtain 4 optimal clusters.

Baseline Model Comparisons

To evaluate the effectiveness of the TFTs performance, it is compared to baseline a baseline LSTM model that was tuned to our purpose of regime classification. To ensure that both models are comparable parameters were set kept to ~70,000. The specifications for each model were as followed:

TFT	LSTM
• Hidden size: 24 • Number of regimes: 4 • Attention Head size: 2 • Hidden Continuous Size: 12 • Learning rate: 0.001 • Optimizer: Ranger • Loss: Cross Entropy (weighted) • Batch size: 64 • Epochs: 20	• Hidden size: 128 • Number of regimes: 4 • Learning rate: 0.001 • Optimizer: Adam • Loss: Cross Entropy (weighted) • Batch Size: 64 • Epochs: 20

Table 3. Parameters of TFT and LSTM models

TFT & LSTM Models

I implement a TFT using the PyTorch Forecasting Framework. The TFT was trained to classify each time step to one of four regimes based on the K-means regime labeling strategy. The model's input variables were as follows:

- Time Varying Known Categorical: numerical days of the week, numerical month, and numerical quarter
- Time Varying Known Reals: Z-scores for macroeconomic indicators (VIX, EFFR, TR30Y, Uncertainty Index, AAA spread, High Yield Spread), S&P 500 return
- Time Varying Unknown Reals: Realized Volatility Z-score, Realized Volatility Lag sequences (1, 2, and 3), S&P 500 Squared log returns, Volume Z-score, Log returns Z-score

I also implemented an LSTM-based classifier in PyTorch. The inputs were as followed:

- Time Varying Known Categorical: numerical days of the week, numerical month, and numerical quarter
- Time Varying Unknown Reals: Realized Volatility Z-score, Realized Volatility Lag sequences (1, 2, and 3), S&P 500 Squared log returns, Volume Z-score, Log returns Z-score, Z-scores for macroeconomic indicators (VIX, EFFR, TR30Y, Uncertainty Index, AAA spread, High Yield Spread), S&P 500 return

Because classification was the main functionality of the model, a weighted Cross Entropy loss function was also used for evaluation of metrics and performance break down.

4 Results

In this paper, I explored the predictive power of the TFT and the LSTM for use in regime classification. The metrics used for comparison were Accuracy, Precision, Recall, and F1 Score.

I. Accuracy – Simplest metric that measures the proportion of correct classifications
   a. $Accuracy = \frac{Correct\ Predictions}{All\ Predicitons}$
II. Precision – Represents the reliability of correctly predicted regimes.
   a. $Precision = \frac{True\ Positive}{True\ Positive + False\ Positive}$
III. Recall – Measures the model's predictive power to detect extraordinary events.
   a. $Recall = \frac{True\ Positive}{True\ Positive + False\ Negative}$
IV. F1 – Combination of precision and recall that calculates harmonic mean – gives greater importance to smaller values.
   a. $F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$

Measurement	Accuracy	Precision	Recall	F1
LSTM	86%	86%	1.00	0.14
TFT	17%	8%	0.50	0.92

Table 4. Metrics for Validation Set for LSTM vs. TFT

The LSTM was 86% accurate in predicting the correct class, however, the TFT was only 17% accurate. Next, LSTMs predicted the correct regime 86% of the time while 8% of the TFTs predictions were true. The recall of the LSTM was perfect because it never missed a true change, however, the TFT caught half of real changes. Lastly, the TFT obtains an F1 score of 0.92 signifying that it can achieve a persistent balance in differentiating real events and avoiding misses. Conversely, the LSTM achieves an F1 score of 0.14 indicating it may be only predicting the most prevalent regime in the dataset.

5 Discussion

I anticipated inputting a rich data set within a Temporal Fusion Transformer would lead to better regime detection than traditional deep learning classification methods. However, the results challenged my hypothesis. Although the LSTM was able to significantly outperform the TFT in most metrics, upon closer inspection the output of the LSTM was a result of safe prediction by outputting the most prevalent regime. Conversely, the TFT significantly underperformed however, was able to achieve a respectable balance between its precision and recall. These metrics may be a result of the sensitivity of the TFT's architecture to the data as well as the bias of the data.

One limitation of the experiment is the mislabeling of regimes from data sources which can lead to missed anomalies. However, this is subjective and highly dependent on the regime detection method deployed. Another restriction was limited computing power which caused restrictions in optimizing parameters of the model.

6 Conclusion

This study aimed to utilize macroeconomic and financial data for integration into a Temporal Fusion Transformer to detect regimes. Through diverse inputs of high and low frequency data gathered, this model was compared to an LSTM. The results of this study indicate that while low predictive accuracy was achieved, the TFT was able to detect structural shifts. Further research can be conducted on utilizing different time-series transformer-models to forecast regimes such as Informer. Furthermore, larger or different data sets may also be beneficial in capturing sound regime and structural shifts in data.

7 References

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8 Appendices

8.A K-means Regime Clustering

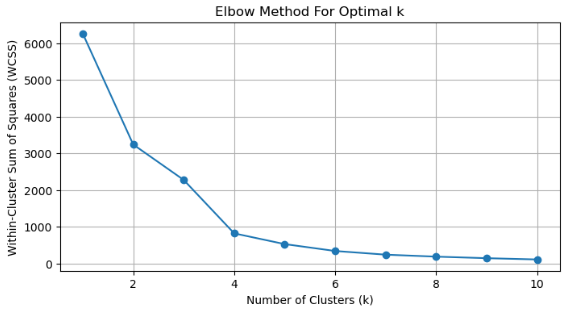

Figure A.1. Elbow method for obtaining optimal number of regimes

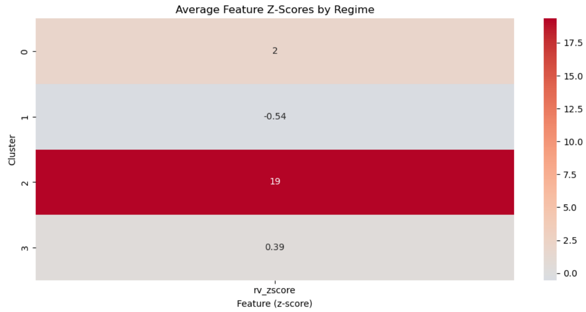

Figure A.2. Heat map of clusters

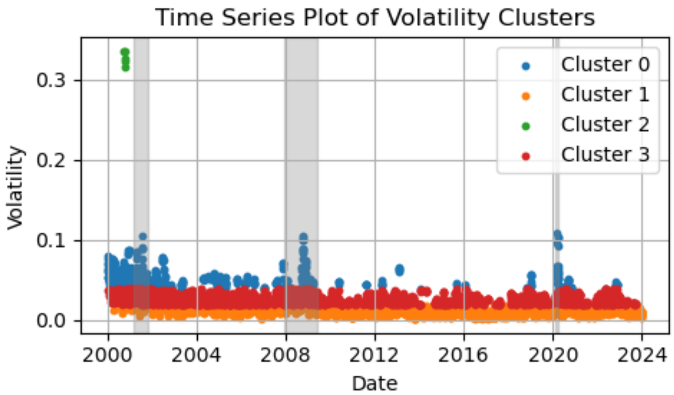

Figure A.3. Volatility Clusters based on Realized Volatility

8.B MS-AR Regime Classification

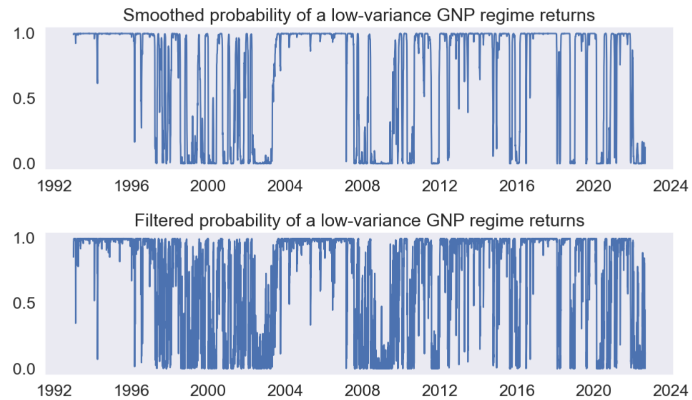

Figure B.1 MS-AR regime probabilities model with SPY returns 1994-2023

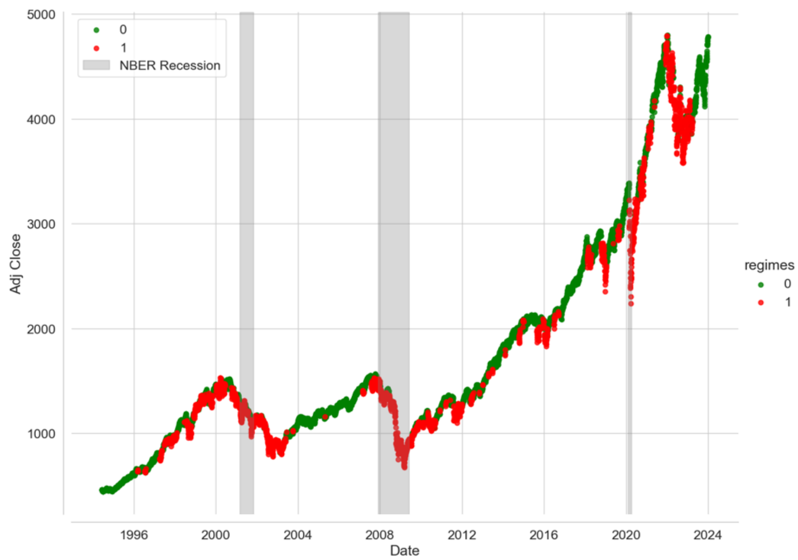

Figure B.2 MS-AR regimes with SPY Close 1994-2023

8.C HMM Regime Classification

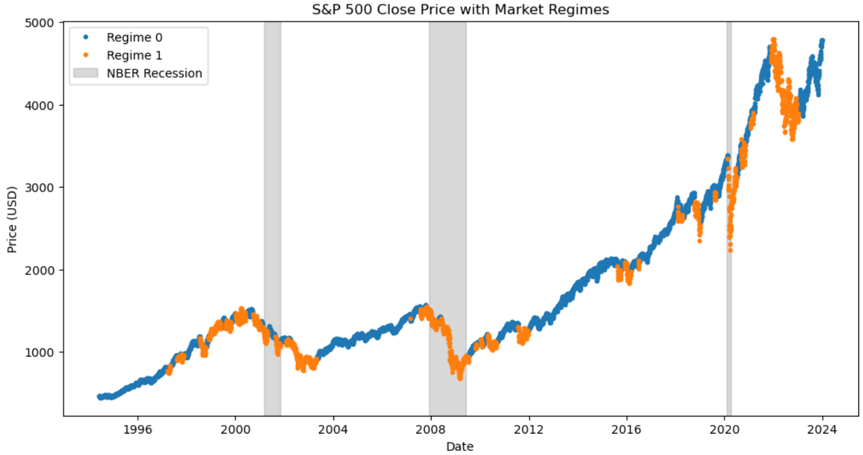

Figure C.1 Hidden Markov Model Regimes with SPY Close with NBER recognized regimes - 1994-2023

8.D Diagrams of results from last 90 days of Validation set

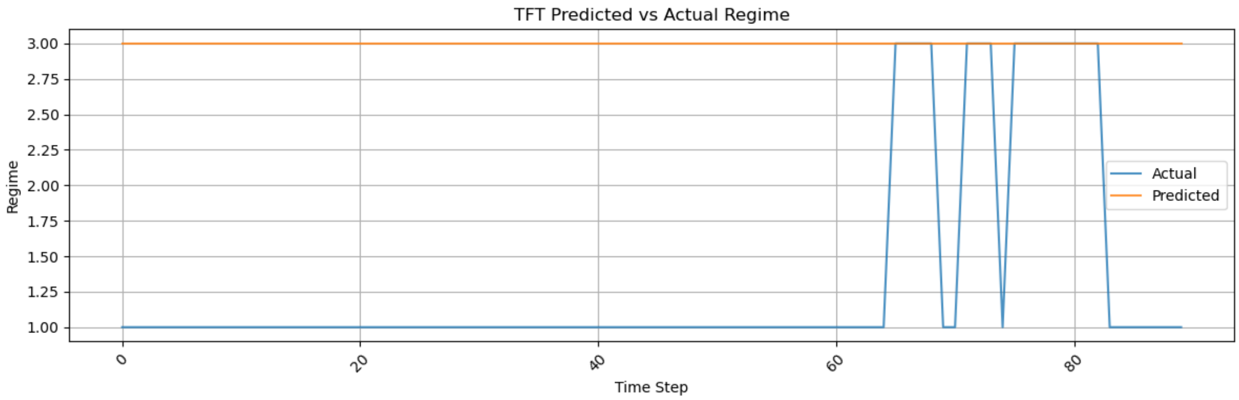

Figure D.1 TFT Predictions vs. Actual in validation set of last 90 time-steps

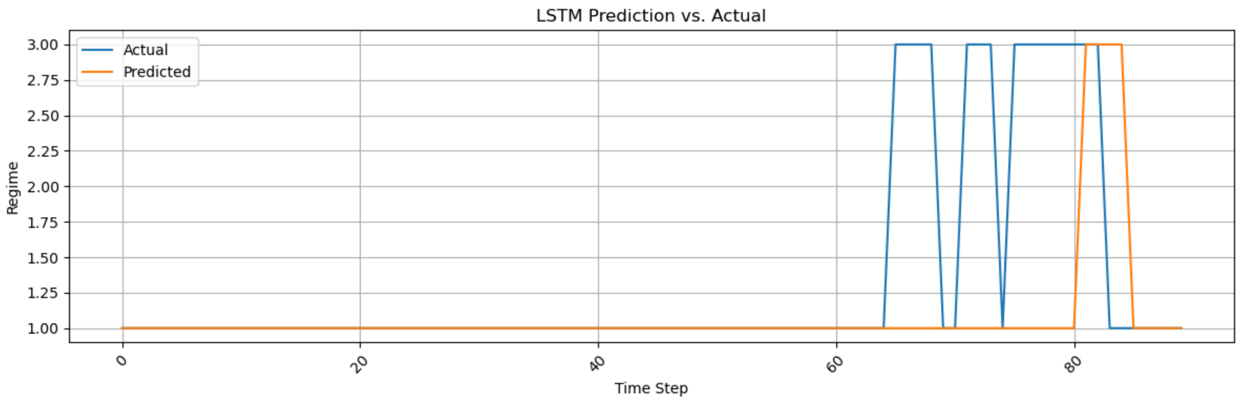

Figure D.2 LSTM Prediction vs. Actual Values in validation set of last 90 time-steps

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